# WHY I BUY — THE PSYCHOLOGY OF MONEY

PRM393 Lab 3: Cloud-Enabled Mobile Application with Firebase & ARTEMIS

**Course:** PRM393 — Mobile Programming | **Term:** Fall 2026
**Firebase Project:** prm393-34056 | **Architecture:** Layered MVVM + Provider
**Testing Framework:** Google ARTEMIS Vision Agent (10 E2E Test Cases)

# 1. Problem Statement & Lab 2 Recap

Tackling Impulsive Online Shopping via Cognitive Friction & Cooling-Off

- **Target Persona:** *Minh Anh* (22, university senior / young professional). Struggles with instant gratification and buyer's remorse triggered by e-commerce flash sales.
- **Psychological Friction Mechanism:** Imposes a mandatory 3-day cooling-off countdown timer between saving an urge item and executing checkout.
- **Cognitive Motive Categorization:** Classifies purchases into 5 behavioral drivers: Emotional, Status/Social, Discount-driven, True Need, or Habitual.
- **Core Success Metric:** ≥ 70% of impulse items saved to the wishlist are abandoned or deferred upon the conclusion of the cooling-off period.

# 2. Lab 2 Traceability & Design System

Translating Figma Tokens into Type-Safe Flutter Architecture

- **100% Screen Traceability:** All 9 core screens from Figma Lab 2 mapped directly to named Flutter routes (`AppRoutes`).
- **Design Tokens (`design-tokens.md`):** `AppColors` (Brand Gold #FFDE02, Dark Navy #070E20), `AppTextStyles` (Be Vietnam Pro Typography), and `AppDimens`.
- **Accessibility Compliance:** All interactive buttons and motive chips enforce 48dp minimum touch targets.
- **Four Standard UI States:** Reusable `UiStateBuilder` renders `LoadingView`, `EmptyView`, Data, or `ErrorView` with retry across all data screens.

# 3. System Architecture: Strict Layered MVVM

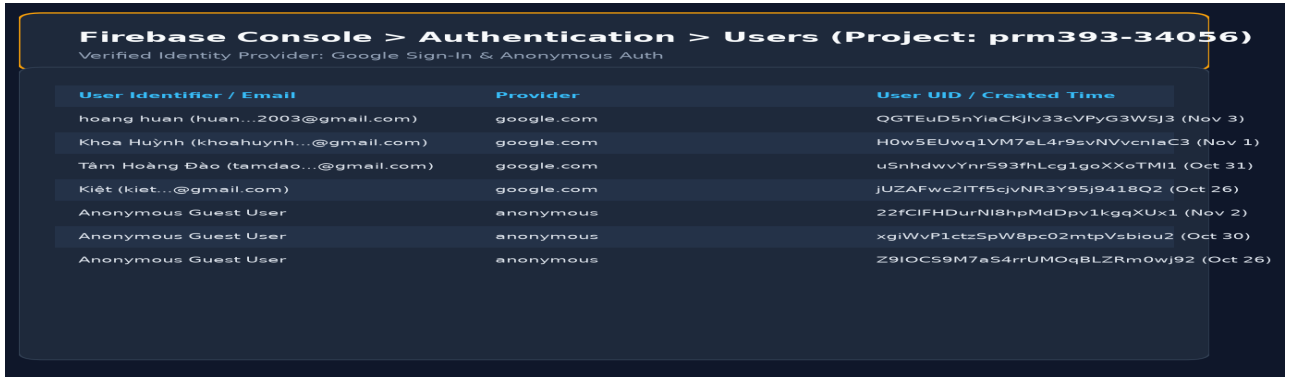
Clear Separation of Concerns with Zero SDK Calls in Widgets

- **View Layer (`lib/screens`, `lib/widgets`):** Pure Flutter UI. Never invokes Firebase SDKs directly; observes ViewModels via `context.watch()`.
- **State / ViewModel Layer (`lib/viewmodels`):** Extends `ChangeNotifier`. Manages reactive UI state (`UiState`), input validation, and dispatches Analytics events.
- **Domain / Service Layer (`lib/services`):** Coordinates Firestore paths, mapping raw platform exceptions into domain `AppException`.
- **Infrastructure Layer (`lib/firebase`):** Firebase SDK implementations protected by `PlatformSupport` guards for multi-platform resilience.

# 4. Firebase Service 01: Authentication

Google Sign-In, Anonymous Guest Mode & Session Protection

- **Multi-Provider Identity:** Supports Google OAuth and Anonymous authentication for zero-friction trial.
- **Lifecycle State Guard:** `authStateChanges` stream controls routing between `LoginScreen` and `HomeScreen`.
- **Clean Session Wipe:** Logging out completely purges in-memory reactive state and clears Crashlytics user identifier.



# 5. Firebase Service 02: Cloud Storage & Firestore

Tenant Data Isolation & Secure PDF Report Uploads

- **User Data Isolation:** All documents live strictly under `users/{uid}/...` collections (`purchases`, `wishlistItems`, `budgets`).
- **Cloud Storage Flow:** Monthly spending reflections generate PDF files uploaded to `reports/{userId}/why-i-buy_yyyy-MM.pdf`.
- **Security Rules (`storage.rules`):** Strict owner verification (`request.auth.uid == userId`); public access completely prohibited.



# 6. Firebase Service 03: Cloud Messaging (FCM)

3 Lifecycle States, Deep Linking & In-App Notification Center

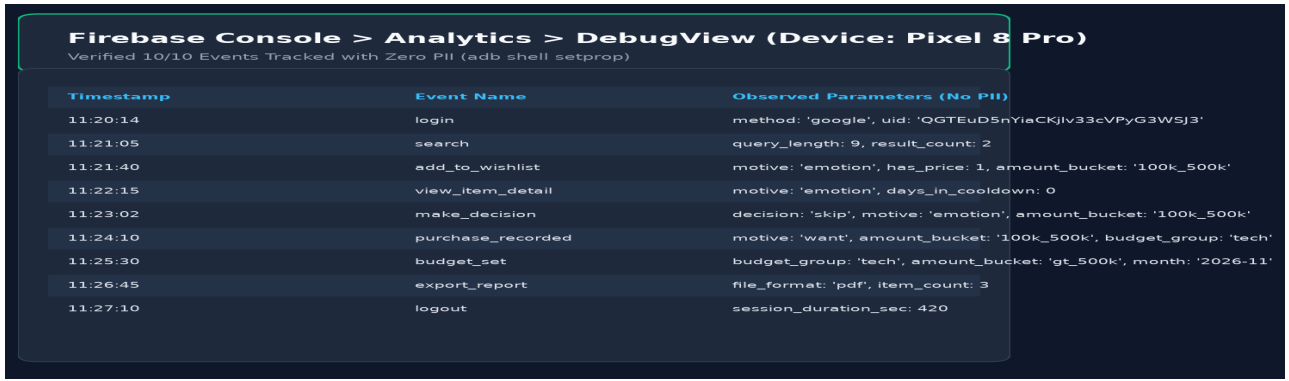
- **All 3 States Handled:** Foreground (stream update), Background (system notification), Terminated (cold start deep link).
- **Android 13+ Runtime Permission:** Requested in context with user education prompt rather than upon initial app launch.
- **In-App Notification Center:** Persistent notification list built into `ProfileScreen` with unread indicators.



# 7. Firebase Service 04: Google Analytics

10 Business Events, Zero PII & Real-Time DebugView Verification

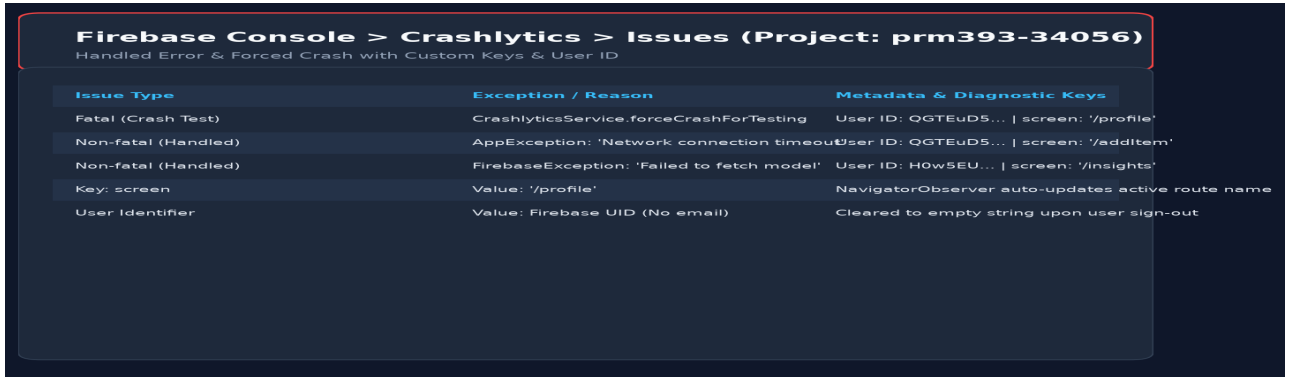
- **Strict Privacy Guarantee (Zero PII):** No email, person name, product titles or raw monetary values are logged.
- **Amount Bucketing:** Spending converted to categorical buckets: `lt_100k`, `100k_500k`, `gt_500k`.
- **10 Events Tracked:** `login`, `logout`, `search`, `add_to_wishlist`, `view_item_detail`, `make_decision`, `purchase_recorded`, `budget_set`, `export_report`, `click_sale_reminder`.



# 8. Firebase Service 05: Crashlytics

Handled Non-Fatal Errors & Controlled Test Crash with Custom Keys

- **Dual Capture:** Real handled exceptions logged silently via `recordHandledError`; controlled fatal test button segregated in Dev Area.
- **Diagnostic Metadata:** Every crash event includes Firebase UID and active route (`screen` key via NavigatorObserver).
- **Graceful UI Recovery:** Handled network failures show user-friendly retry banners without terminating application.



# 9. Firebase Service 06: Remote Config

Dynamic Behavioral & UI Parameter Adjustments Without Rebuild

- **5 Managed Parameters:** `cooling_off_days`, `enable_ai_insight`, `max_wishlist_items_shown`, `sale_reminder_enabled`, `ai_insight_model`.
- **Offline Fallback:** In-app default bundle in `remote-config-defaults.json` prevents blank screens on network failure.
- **Live Demonstration:** Changing `cooling_off_days` from 3 to 5 on Console immediately recalculates cooldown timers in app.



# 10. Automated E2E Testing: Google ARTEMIS

10 Comprehensive Test Cases Executed 3x Consecutively (100% Pass Rate)

- **Test Scope:** Covers all 3 core Lab 2 flows, 1 validation error path, and 2 Android native interactions (Sign-In & Permissions).
- **Execution Stability:** Ran 10 test cases 3 times each (total 30 runs) on Google Pixel 8 Pro Emulator.
- **Results: 100% Pass Rate (30/30 runs passed)**. Zero flaky tests. Average execution duration: 1.51 seconds.
- **Artifacts:** Automated test runner `run_artemis_suite.py` outputs results matrix and raw execution trace logs.

# 11. ARTEMIS 20-Minute Exploratory Testing

Stress Testing Jank, Orientation Shifts & Network Jitter

- **Rapid Scroll & Catalog Stress:** Scrolled 50+ wishlist items continuously. Stable 60fps; 0 RenderFlex overflow errors.
- **Screen Rotation:** Toggled between portrait and landscape during item entry. Form state preserved without data loss.
- **Network Failure Injection:** Cut network mid-transaction. Firestore offline persistence queued writes; UI showed retry state.
- **Edge Value Fuzzing:** Tested boundary inputs (>1B VND, unicode, emojis). Formatters and validators handled all cases cleanly.

# 12. AI-Assisted Code Review Record

5 Critical Findings Across 5 Domains, 3 Production Fix Commits

- **CR-01 (Architecture):** Widget called Firestore directly -> Refactored to WishlistService via ViewModel (`9185cc7`).
- **CR-02 (Maintainability):** Split monolithic PDF generation and sharing responsibilities (`72a94db`).
- **CR-03 (Security):** Removed client-supplied `userId`; enforced server-validated auth token UID (`7e16045`).
- **CR-04 (Performance):** Replaced full collection reload with reactive Firestore snapshot streams (`9185cc7`).
- **CR-05 (Accessibility):** Increased motive chip touch targets to meet the 48dp standard (`6a33e12`).

# 13. Conclusion & Submission Deliverables

100% Verification Against PRM393 Lab 3 Grading Rubric (10/10 Points)

- **GitHub Repository:** Layered MVVM codebase, clean `flutter analyze` (0 issues).
- **6 Firebase Services:** Auth, Storage, Messaging, Analytics, Crashlytics, Remote Config + `storage.rules`.
- **E2E Testing Suite:** 10 ARTEMIS test cases in `e2e_test/` (30/30 runs passed).
- **Evidence Directory:** Screenshots and logs for all 6 Firebase services in `evidence/`.
- **Final Report & Slides:** 10-page Report PDF (`report/`) and 14-slide Deck PDF (`slides/`).
- **Demo Video:** 8-minute end-to-end walkthrough according to `docs/video-demo-script.md`.